

Skin in fishes



SKIN - INTEGUMENT

- Integument is boundary between the organism and the environment.
- Fish like other vertebrates have two types of skin.
 - ✓ **Outer epidermis.**
 - ✓ **Inner dermis.**



STRUCTURE OF SKIN

- The skin of fish contains an epidermis , dermis along with various other structures present on their skin.

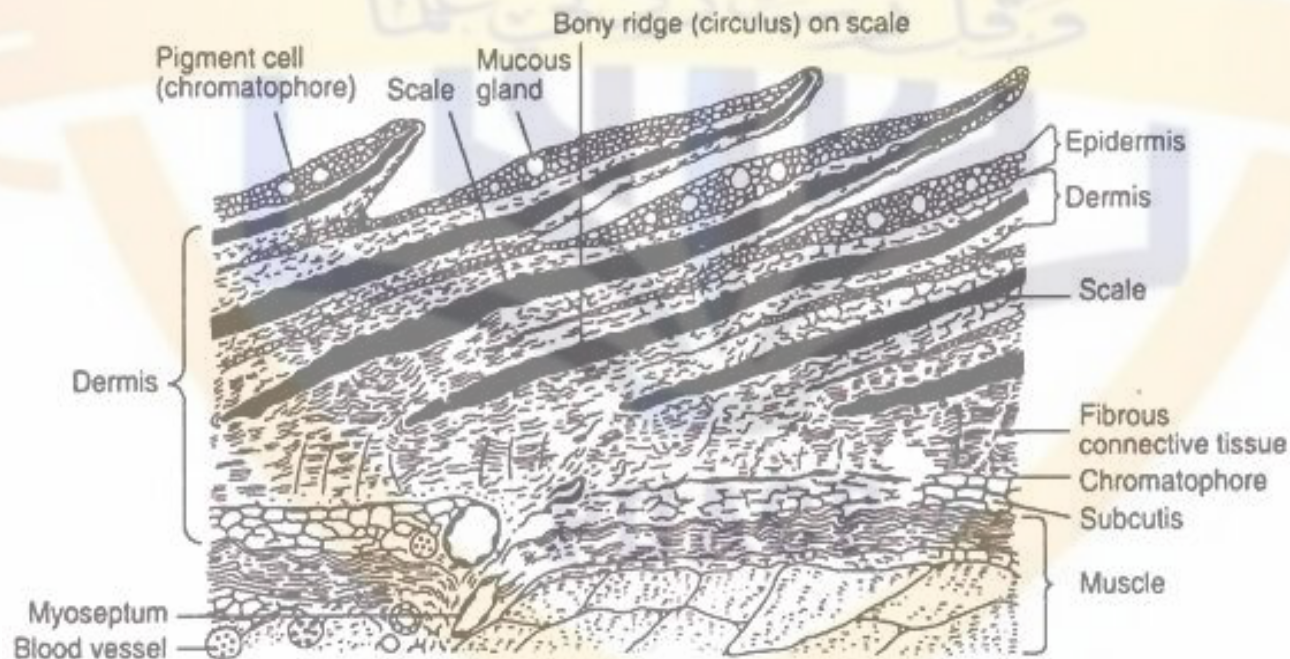
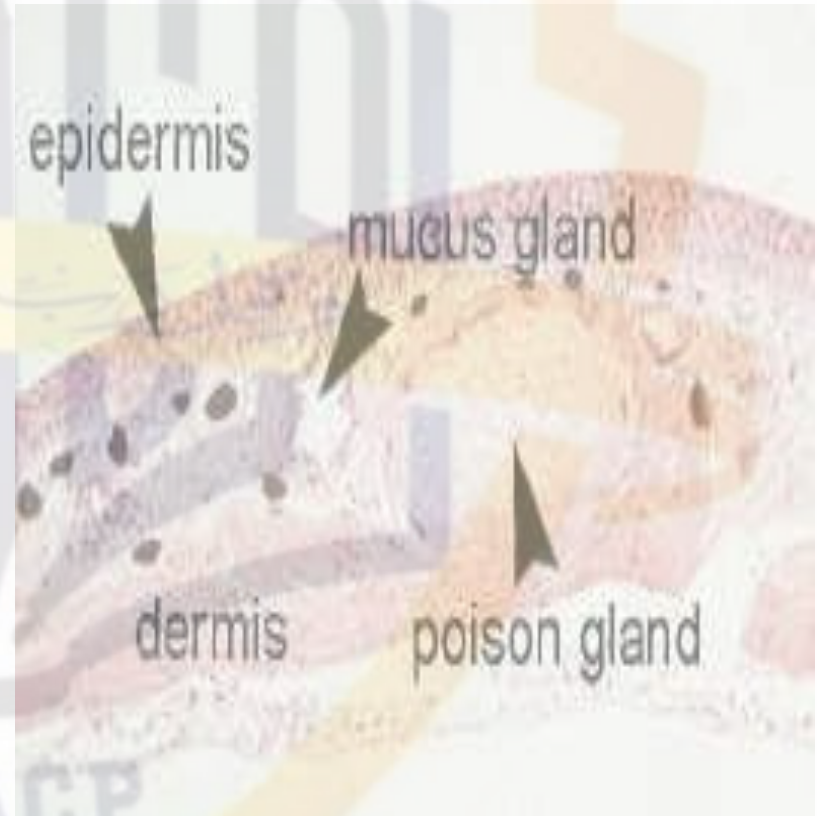


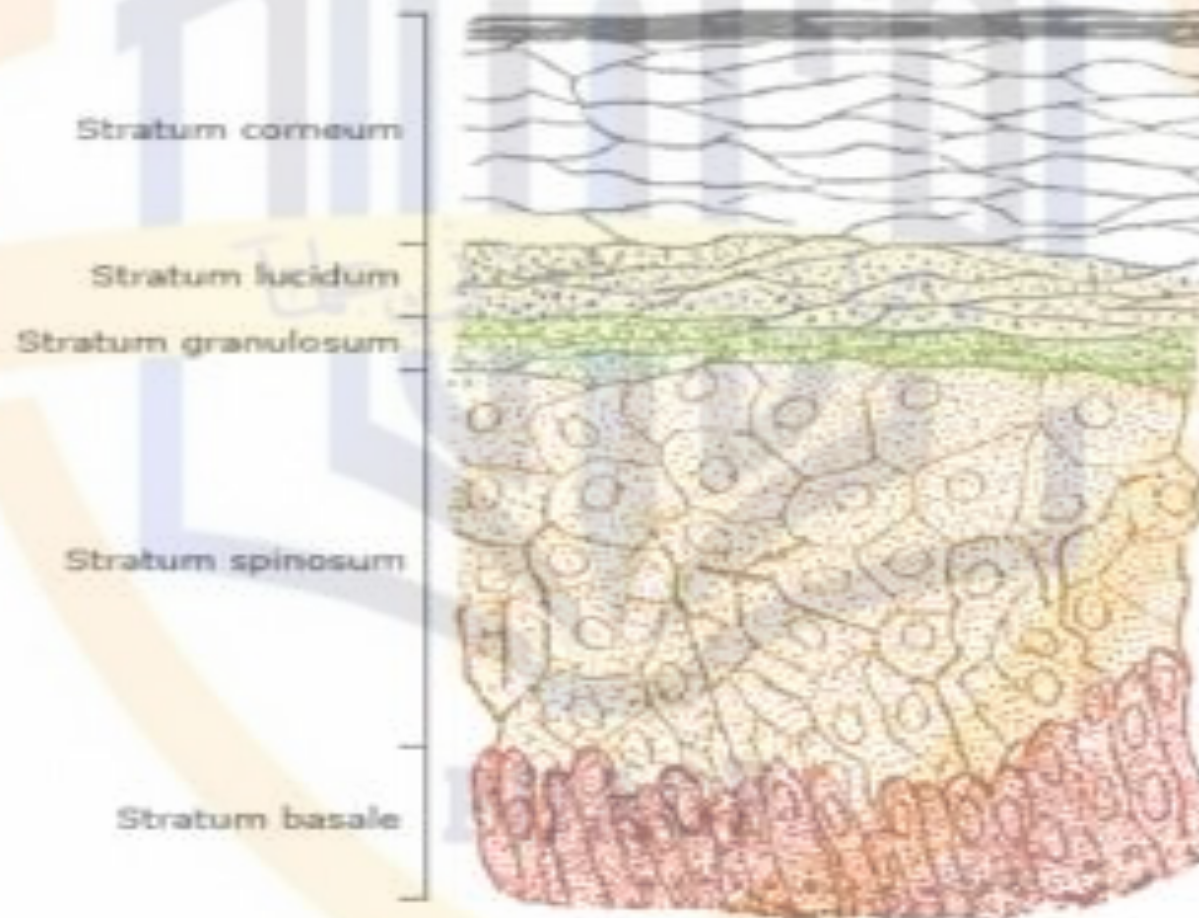
Fig. 3.1 : Section of fish skin (Source : General Biological Supply House)

EPIDERMIS

- Ectodermal in origin and replacement is continuous.
- It is avascular contains no blood vessels.
- Contains mucous cells.
- Mucous cells secrete mucus.



FIVE LAYERS OF EPIDERMIS



FIVE LAYERS OF EPIDERMIS

- **Stratum corneum** layer which contains dead, horny , **keratinized squamous cells** are found only in bony fishes and minimizes the water loss.
- **Stratum lucidum** is thin clear layer of dead cells.
- **Stratum granulosum** is a thin layer of **squamous cells often with lipids** which makes this layer waterproof.

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- **Stratum spinosum** is a multilayered arrangement of cuboidal cells that sits below the stratum granulosum and provides **strength and flexibility to the skin.**
 - **Stratum germinativum** (Stratum Basale) is a basal layer above the dermis .It consists of single layer of column like epithelial cells. These cells undergo rapid divisions , also known as **generating layer of epidermis.**

FUNCTION OF EPIDERMIS

- Slime that is produced is highly protective.
- In lung fishes, the epidermis has a extracellular spongy cuticle to assist in adhesion on rough substrates.

DERMIS

- Mesodermal in origin.
- Thicker than epidermis.
- Blood vessels, nerves , sense organs , connective tissues and pigmented cells (chromatophores) are present in the dermis.

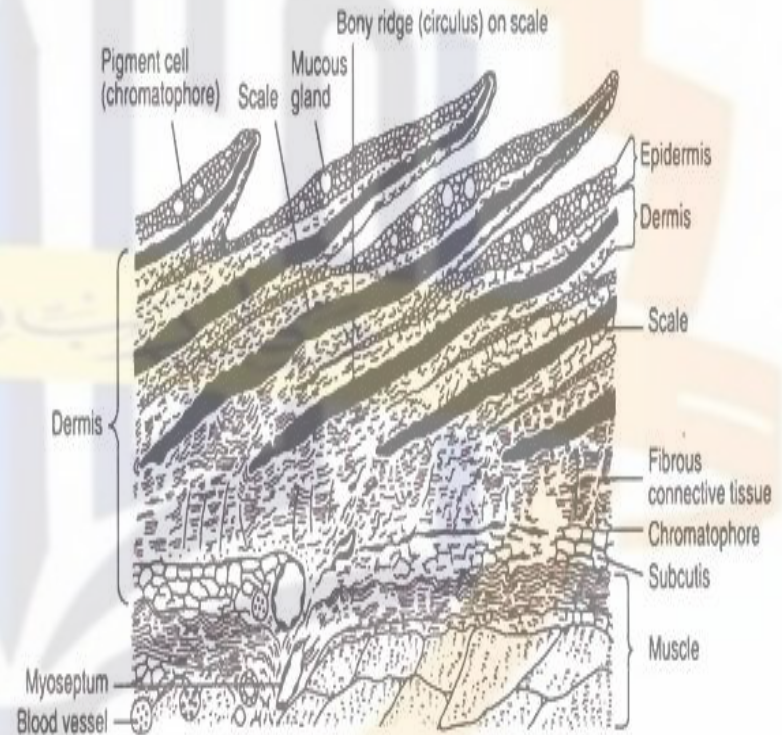


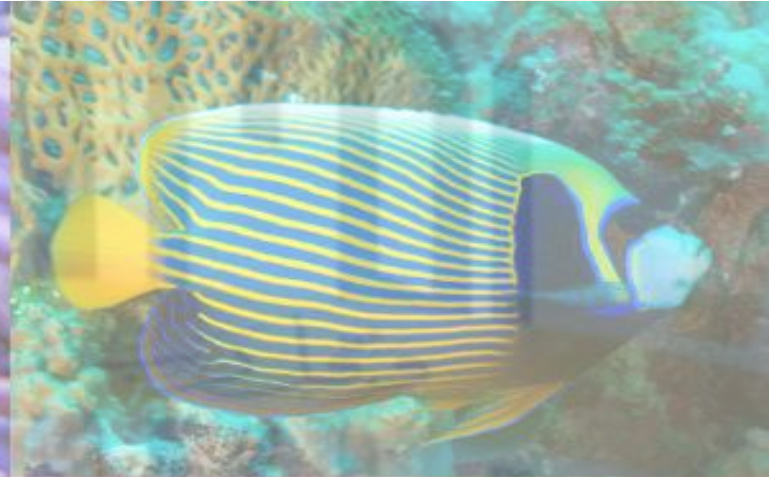
Fig. 3.1 : Section of fish skin (Source : General Biological Supply House)

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- Consists of **two layers** as:
 - ✓ Upper looser layer **Stratum spongioso** and scales are embedded with their bases in this layer.
 - ✓ Lower tighter layer **Stratum compactum** which contains lipid cells and connective tissues .

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- Different pigmented cells (chromatophores) are present having different functions as follows:

- ✓ Melanophores - Black
- ✓ Erythrophores - Red
- ✓ Xanthaphores - Yellow
- ✓ Cyanophores - Blue
- ✓ Leucophores - Light scattering
- ✓ Iridiophores - Light reflecting –silver

FISH COLOURATION



FUNCTIONS OF FISH SKIN

- It protects and separates fish from external environment and harsh conditions.
- Large organ and is continuous with linings of all body organs and also cover the fins.
- Fish integument is a multicellular organ and serve as important roles in protection , excretion , locomotion, respiration , ion regulation and thermal regulation.

SKIN OF JAWLESS FISHES

They have relatively thick skin

Of the several types of epidermal glandular cells, one secretes the protective cuticle

Multicellular slime glands secrete large amount of mucous to cover the body surface for protection



SKIN OF CARTILAGINOUS FISHES

- Multilayered and contains mucus and sensory cells
- The dermis contains bones in the form of placoid scales called denticles
 - **COOKIE-CUTTER SHARK**
 - MANTA RAY** Denticles contain blood vessels and nerves and is very familiar to vertebrate teeth



SKIN OF bony FISHES

- They contain scales
- A thin layer of dermal tissue overlaid the superficial epidermis normally covers the scales
- Because scales are not shed, they grow at the margins and over the lower surface
- Their skin are permeable and can be used in gas exchange
- Mucus are at help in fighting off bacterial and fungal infection at the surface



SCALES IN FISHES!!!

- A scale is a small rigid plate that grows out of an animal's skin to provide protection.
- The study of scale is called **Squamatology**.
- They are usually located between the epidermis and underlying tissues.
- They usually grow on a fish in overlapping tissues, with free end of the scale pointed towards the tail.

TYPES OF FISH SCALES

- The main kinds of scales found in fishes are:
- **Placoid scales.**
- **Ganoid scales.**
- **Bony ridge**
 - **Ctenoid scales.**
 - **Cycloid scales.**

Types of Scales

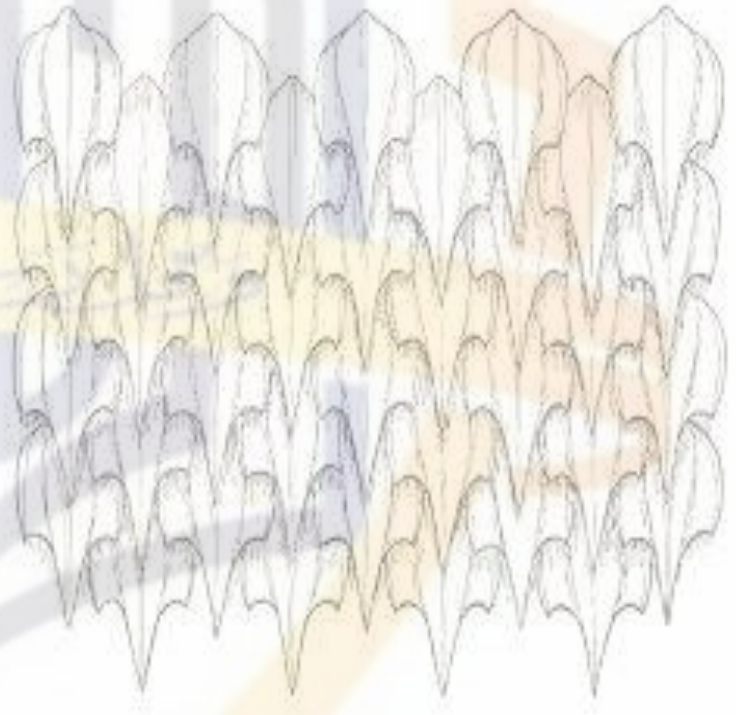
1. Placoid: rectangular base, under the skin, spines project to the surface of the skin

- New scales are added, do not grow with the fish
- Examples: sharks, skates, rays



PLACOID SCALES

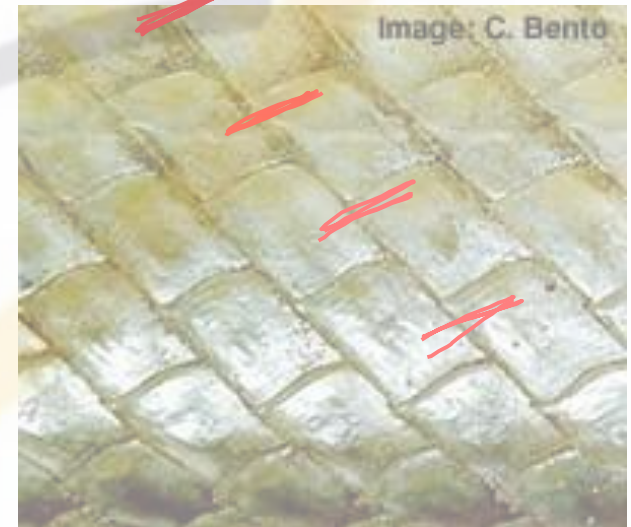
- Similar in structure to teeth, also called denticles.
- Formed of rectangular base plate embedded within the skin.
- The enamel cap called ectodermal ridge is present and without it sharks do not feed.
- Found in **Sharks, skates and rays.**



Types of Scales

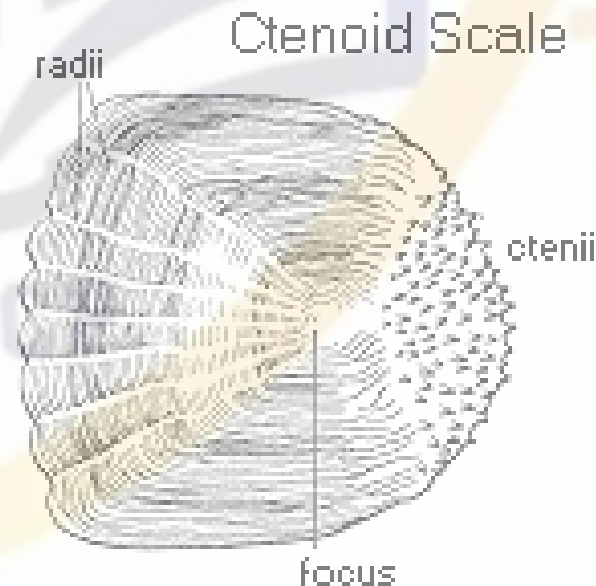
2. Ganoid: rhombus shape, connected by peg and socket joints

- Do not grow with fish, new ones are added
- Examples: Gar, Paddlefish, Sturgeon



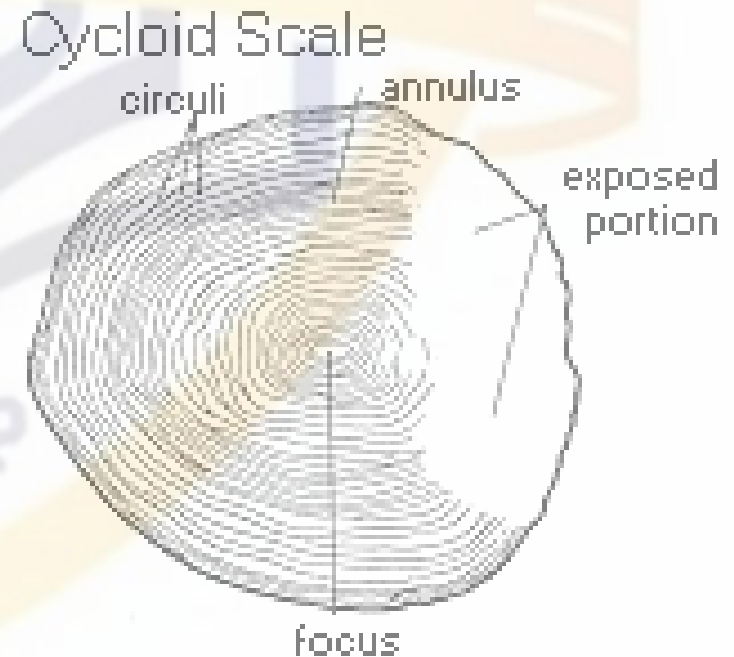
Types of Scales

3. Ctenoid: overlapping, allow great maneuverability as fish bends
- Grows with fish, found in spiny rayed fish



Types of Scales

4. Cycloid: same as ctenoid functions
- Smooth margin, soft-rayed fish
 - Grows with fish



Functions of scales

- Provide protection to its body as scales are tough and slippery.
- Helps to protect fish from parasites and from external injuries.
- Scales provide flexibility so that the scales can slide over each other as fish body bends while swimming.